

Dang Kim Minh Nguyen

• West Lafayette, IN 47906 • (317) 618 1739 • minhkim1811@gmail.com • LinkedIn: minh-nguyen01 • Github: minhnguyen18

EDUCATION

Purdue University

West Lafayette, IN

Bachelor of Science in Computer Information and Technology

August 2023 – December 2026

Relevant Course: Object-Oriented Programming, System Programming, Database, Network Engineering, Cryptography, Information

Technology Architecture, Cyber Forensics, Advanced Security Coding

Honors/Award: 1st place in Purdue Design and Innovation Spring 2025 Challenge (\$5000 award) + 1st place in Crowd Favorite Project (\$500 award)

EXPERIENCE

FPT Software Richardson, TX

Full-stack Software Engineer Intern

May 2025 – August 2025

- Delivered customer-facing Android functionality across frontend, backend, and database layers by developing Kotlin and Java components with MVP architecture and object-oriented design.
- Increased API reliability across real-time application workflows by integrating REST endpoints, validating and transforming complex JSON payloads, and implementing structured error handling and logging.
- Improved release quality by validating features against technical requirements, supporting unit and integration testing, and documenting API behavior, design decisions, and troubleshooting procedures.
- Accelerated delivery in an Agile engineering team by contributing to planning, code reviews, debugging, backend integration, and application performance analysis.

NTT Research Inc. Sunnyvale, CA

Data Science Intern

June 2024 – August 2024

- Achieved approximately 80% forecasting accuracy by developing and evaluating a Python time-series model with Pandas, NumPy, feature engineering, and large-scale SQL datasets.
- Enabled real-time energy optimization by deploying forecasting logic through a production REST API and integrating model outputs with an operational energy-management system.
- Improved data reliability by building reusable Python workflows to profile, clean, validate, and transform structured datasets before model training and evaluation.
- Supported data-driven engineering decisions by analyzing model performance, troubleshooting data-quality issues, and presenting dashboards and technical findings to cross-functional stakeholders.

PROJECTS

AgentForge - AI Agent Platform | React, TypeScript, FastAPI, Python, MySQL, ChromaDB, Ollama, Qwen3, Docker

- Engineered a local-first autonomous AI agent with a custom orchestration loop that plans multi-step tasks, selects and executes tools, evaluates intermediate results, and iterates to completion without relying on paid LLM APIs.
- Built sandboxed Python execution, file operations, web research, and long-term vector memory, enabling the agent to combine tool use with persistent context across complex workflows.
- Designed fault-tolerant agent execution with retry logic, iteration limits, and circuit breakers, achieving 86.7% task completion and 80% failure-recovery across 30 evaluation tasks while preventing infinite retry loops and repeated tool failures.

StudyAI - Real Time Study Platform | React, TypeScript, Node.js, Express, MySQL, Socket.io

- Unified six learning workflows collaborative notes, AI tutoring, flashcards, quizzes, progress tracking, and study recommendations by building a full-stack platform with REST APIs and relational data models.
- Enabled real-time collaboration by implementing live editing, user presence, and typing indicators with Socket.io event validation, logging, and fault-tolerant error handling.
- Automated content-grounded explanations and assessments by building a RAG pipeline that retrieved source chunks from uploaded materials and supplied them to prompt-based AI workflows.
- Improved AI output traceability by storing retrieved chunk references, generated responses, and evaluation data for debugging, validation, and response-quality analysis.

Nautilus - ASL Learning App | Python, OpenCV, MediaPipe, Tensorflow, PySide6 | Winner Project in Purdue Design and Innovation Challenge

- Delivered an award-winning application on schedule by leading a cross-functional team of five and coordinating AI-model, user-interface, testing, and integration work.
- Integrated live camera input, AI inference, and a PySide6 interface into an end-to-end application by profiling latency and debugging model-to-UI bottlenecks.
- Improved real-time usability by profiling application behavior, debugging model and interface bottlenecks, and integrating camera input with AI inference and the desktop user interface.

SKILLS

Languages: Python, C/C++, Java, JavaScript, TypeScript, Kotlin, SQL

Frameworks & Tools: React, Node.js, Express, Django, OpenCV,

MediaPipe, PySide6, Linux

Cloud & DevOps: AWS (IAM, Config), Kubernetes, Firewall Rules

Databases: SPSS, MySQL, PostgreSQL

Machine Learning: Scikit-learn, Numpy, Matplotlib

Security: SSL/TLS, OpenSSL, AES-256, RSA-2048, bcrypt, buffer overflow protection